Venkata Lakshmi Parimala Pasupuleti

IT Application Integration Engineer · Enterprise Systems Integration · Database & Middleware · Multi-Platform APIs & Legacy Systems · 9yr Production

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PROFESSIONAL SUMMARY

IT application integration engineer with 9 years owning enterprise system integrations, multi-platform data pipelines, and compliance-critical application workflows in production environments. At Entrata, I was the long-term owner of integrations between internal CRM platforms and external third-party vendor applications, the architect of legacy system modernization projects (migrating hundreds of thousands of records to new database schemas while keeping production systems live), and the sole owner of regulatory data import pipelines processing millions of records annually. I gather business requirements from end users, design component and data architecture, make buy-vs-build recommendations, and deliver full project lifecycle -- from requirements through implementation, testing, and post-deployment stabilization. I have built HIPAA-compliant health data integration platforms and understand the compliance and security constraints that govern clinical data systems. Currently a Research Associate at George Mason University (M.S. Data Analytics Engineering, GPA 3.87, May 2026) -- working inside an academic institution reinforces my understanding of the stakeholder environment and data stewardship expectations this role requires. Portfolio: venkatalakshmiparimala.github.io/Portfolio/about.html

TECHNICAL SKILLS

Integration & Systems Architecture: Application Integration, System Integration, Multi-Platform Integration, Legacy System Modernization, Middleware Technologies, API Integration, REST APIs, Web Interface Integration, Data Exchange Standards, Enterprise Application Architecture, Component Architecture Design, Data Architecture Design, Buy vs. Build Evaluation, Software Product Evaluation, Requirements Analysis, Implementation, Testing & Customization, System Documentation & SOPs

Database & Data Engineering: SQL/PL-SQL (9yr production), PostgreSQL (200+ query optimization, stored procedures, complex schemas, multi-tenant environments), Oracle (familiarity; PL/SQL transferable), MySQL, Microsoft SQL Server (T-SQL), ETL/ELT Pipelines, Data Modeling, Schema Design, Data Validation & Quality Controls, Data Governance, Database Performance Engineering, Deduplication, Data Cleansing

Development, Project & Infrastructure: Python (9yr primary), JavaScript, HTML/CSS, PHP (familiarity), XML, Bash scripting, AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), Git, Agile/Scrum, Project Management, Stakeholder Communication, End-User Requirements Gathering, Reporting & Dashboards, Tableau, Power BI; Security & Access Controls, Data Privacy, HIPAA-compliant systems

EXPERIENCE

Entrata -- Multifamily Real Estate SaaS (PropTech)

Sep 2018 -- Aug 2024

Senior Software Engineer -- Enterprise Systems Integration, Application Development & Compliance Workflows · Lehi, UT (Remote)

- **Multi-platform application integration and third-party data synchronization** -- long-term owner of integrations between internal CRM and platform systems and external vendor applications; managed data import pipelines from multiple third-party sources into a multi-tenant enterprise system; defined data exchange standards, monitoring protocols, alerting, and error-resolution workflows; coordinated with operations and business teams to gather integration requirements and deliver implementations with zero disruption to production users. AWS S3 migration with encrypted ETL pipelines (100% PHI compliance, zero security incidents).
- **Legacy system integration and application modernization** -- led integration of federal regulatory data systems (HUD income-limit pipeline: thousands of source records transformed and mapped to ~180,000 domain records) into Entrata's proprietary platform annually; led database schema redesign migrating hundreds of thousands of records to a modernized architecture with 6 months of post-deployment stabilization; evaluated buy-vs-build decisions for new integration components across the affordable housing module lifecycle.
- **Database and middleware engineering** -- optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, partitioning, and stored procedure refactoring; designed and maintained complex multi-tenant schemas for eligibility, financial, and document workflows; built RabbitMQ-based async middleware for gross-rent processing across thousands of tenant records with idempotency guarantees and retry logic eliminating double-processing failures.
- **Project management and stakeholder delivery** -- owned 20+ full-stack application modules end-to-end across 120+ Agile sprints; consulted with operations, compliance, and business leadership to determine requirements, set priorities, and manage delivery scope; built reporting and dashboard frameworks for ad-hoc and scheduled analytics delivery; maintained 99.5% uptime; mentored 5+ engineers and conducted 300+ peer code reviews.

George Mason University

Aug 2024 -- May 2026

Research Associate & Graduate Teaching Assistant

- Built multi-source data pipelines and prediction models (Python, R) across 190+ countries (R^2=0.977); applied Isolation Forest to automate data integrity checks across 3 research pipelines; designed Tableau and ggplot2 dashboards improving research comprehension 40% for faculty leadership; delivered data analytics instruction to 600+ students.

Magizhs Tech Solution

May 2025 -- Aug 2025

Engineer -- Data Systems & Multi-Step API Integration

- Designed and built multi-step data retrieval and validation pipelines with conditional routing across external API integrations; implemented data transformation and quality rules; established automated validation scripts enforcing data integrity before export.

Entrata India -- Pune

Jan 2017 -- Sep 2018

Associate Software Engineer -- Application Development & Systems Integration

- Built compliance data modules and application integration workflows; income-limit import pipeline (1.6M HUD/Tax Credit records); AWS S3 migration resolving 90% of legacy storage constraints; 56 state-specific document templates; database schema redesign with post-deployment stabilization.

PROJECTS

AI on FHIR: HIPAA-Compliant Health Data Integration Platform

Flask · REST API · HIPAA · RBAC · Health Data Integration · Full Stack

Python · Flask · React · Docker · FHIR API · spaCy · BERT | github.com/venkatalakshmiparimala/AI_on_FHIR

HIPAA-compliant health data integration platform: REST API integration with external FHIR-standard health data systems, RBAC access controls, secure retrieval across structured clinical records, containerized deployment. Demonstrates integration architecture for regulated health data including electronic health records and clinical data systems -- the same compliance and data security requirements UMSOD's clinical and radiology systems demand.

AgentProbe: Automated Integration Testing & Validation Framework

Python · FastAPI · REST API · CI/CD · Automated Testing · Reporting

Python · FastAPI · React.js · TypeScript · GitHub Actions · PostgreSQL | github.com/venkatalakshmiparimala/AgentProbe

Automated integration validation framework: 5-metric quality engine tracking correctness and consistency across data exchange pipelines; CI-integrated regression detection alerting on integration failures before they reach production; REST API backend with reporting dashboards for stakeholder analysis. Built to enforce integration reliability -- the same discipline needed for production clinical and radiology system integrations.

CrashClarity: Large-Scale Data Pipeline & Reporting (104K Records)

Python · R · Tableau · Data Analysis · Reporting · Dashboard

Python · R · scikit-learn · Random Forest · Tableau · GitHub Pages | Portfolio

End-to-end data pipeline processing 104K+ records -- cleaning, transformation, feature engineering, predictive modeling (85%+ accuracy), and interactive Tableau dashboard deployed for public access. Demonstrates full pipeline ownership from raw source data to stakeholder-ready reporting.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2013--2017